SHUBHAM SINGH

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EDUCATION

Bachelor of Technology (B.Tech) — CSE (IoT with Cybersecurity in Blockchain Technology) 2023 – 2027 (Expected)
Lakshmi Narain College of Technology, Bhopal (M.P.)
CGPA: 7.33 / 10.0

Senior Secondary (XII) — CBSE 2022
St. John's Sr. Sec. School, Damoh (M.P.)
Percentage: 66.7%

Secondary (X) — CBSE 2020
St. John's Sr. Sec. School, Damoh (M.P.)
Percentage: 84%

TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced), C++ (Intermediate), SQL
- **Web Development:** HTML, CSS, JavaScript, Django, REST APIs, AWS (Basic)
- **Frameworks & Libraries:** Django REST Framework, React.js, Tkinter, OpenCV, Matplotlib, Pandas
- **AI & Data:** Google Gemini API Integration, PDF Text Extraction (pdfplumber), SM-2 Spaced Repetition Algorithm
- **Databases:** MySQL, SQLite, PostgreSQL (Neon), Database Design & ORM
- **IoT & Embedded Systems:** ATMEGA328P, Arduino, LiDAR, IMU Sensor Fusion, nRF24L01 Wireless, GPS Modules
- **Tools & Platforms:** Git & GitHub, Salesforce, ServiceNow, Kali Linux, Blender, Fusion 360, Vite, Vercel, Render
- **CS Fundamentals:** Data Structures & Algorithms, Computer Networks, Blockchain, Cybersecurity, Operating Systems

PROJECTS

Projectile Reconnaissance Device (PRD) | *IoT Hardware + Embedded C + React Dashboard* Nov 2025 – Present
GitHub: github.com/yash-goyal-0910/Projectile-Recon-Device

- Designed and led development of a low-cost, disposable IoT reconnaissance capsule (~Rs.10,000) for mapping and environmental sensing in hazardous, inaccessible locations such as collapsed buildings, mines, and caves.
- Engineered multi-sensor data fusion system integrating LiDAR-Lite v3 (360-degree 2D/3D mapping), BNO055 IMU (absolute orientation), BME680 (temperature, humidity, pressure, air quality), AMG8833 IR thermal grid (8x8 heatmap), and Adafruit Ultimate GPS (66-channel, 10 Hz).
- Invented a computational leveling algorithm using rotation matrices derived from IMU orientation data to digitally correct LiDAR point-cloud data for arbitrary tilt angles, eliminating costly mechanical stabilizers.
- Implemented wireless data transmission via nRF24L01 (2.4 GHz, 250 Kbps) with CRC checksum encoding to a remote Ground Control Station for real-time processing.
- Built a React.js-based Mission Control Dashboard featuring real-time IMU gauges, LiDAR sweep visualization, thermal heatmaps, and environmental sensor charts.
- Created 3D model in Fusion 360, circuit schematics, and drop-simulation in Blender; firmware prototype in Embedded C on ATMEGA328P microcontroller.

  Tech Stack: Embedded C, C++, React.js, Arduino, LiDAR, IMU, nRF24L01, GPS, BME680, AMG8833, Fusion 360, Blender, Circuit.io

AI Personal Study Assistant | *Full-Stack Web Application (Django + React)* May 2026 – Present

GitHub: github.com/ShuBhaMSinGhSama/AI-Personal_stdy_Assistant

- Developed a full-stack AI-powered study companion web application with Django REST Framework backend and React 19 + Vite frontend, deployed on Render (backend) and Vercel (frontend).
- Integrated Google Gemini 2.0 Flash API for intelligent AI chat tutoring with conversation history context (last 20 messages) and study material context injection (up to 8,000 characters).
- Built an AI-driven flashcard generation system that automatically creates structured flashcards from uploaded study materials using Gemini API with robust JSON parsing and validation.
- Implemented SM-2 Spaced Repetition Algorithm for intelligent review scheduling, tracking easiness factor, interval days, repetitions, and next review dates to optimize long-term retention.
- Engineered PDF upload and processing pipeline using pdfplumber for automatic text extraction, enabling AI-powered content analysis of uploaded study documents.
- Designed JWT-based authentication system (SimpleJWT) with 24-hour access tokens, 7-day refresh token rotation, token blacklisting, and per-user data isolation across all endpoints.
- Created a responsive dark glassmorphism-themed UI with 8 pages: Dashboard (analytics), Chat (AI tutor), Materials (CRUD + PDF upload), Flashcards (CRUD + AI generation), Review Mode (SM-2), Sessions (tracking), Login, and Register.
- Built RESTful API with 4 Django models (UUID PKs), 4 ViewSets, 4 custom endpoints, and pagination — backed by SQLite (dev) and PostgreSQL via Neon (production).

  Tech Stack: Python, Django, Django REST Framework, React.js, Vite, JavaScript, Google Gemini API, JWT Auth, PostgreSQL, SQLite, pdfplumber, Vercel, Render, Git

CERTIFICATIONS

Python Essentials (1 & 2) — Cisco Networking Academy 2025

Ethical Hacker — Cisco Networking Academy 2025

Networking Essentials — Cisco Networking Academy 2025

Certified System Administrator (CSA) — ServiceNow 2026

DOMAINS & INTERESTS

- Internet of Things (IoT), Embedded Systems, Sensor Fusion, Wireless Communication Protocols
- Artificial Intelligence, API-based Projects, GUI & CLI Application Development
- Cybersecurity, Penetration Testing Tools (Basic), Blockchain Technology